

MUNDARE AGDM, AB, Canada**WMO: 713540****Lat: 53.5606N****Lon: 112.2961W****Elev: 690****StdP: 93.31****Time Zone: -7.00 (NAM)****Period: 03-19****WBAN: 99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.2	-27.3	-34.0	0.2	-30.0	-30.8	0.2	-27.2	12.0	-0.7	10.7	-1.9	3.1	220	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	28.0	17.1	26.2	16.4	24.5	15.8	19.2	25.4	18.0	24.1	16.9	22.4	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.1	22.5	15.7	12.1	20.8	14.6	11.3	19.4	57.7	25.3	53.6	24.2	50.0	22.3	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.1	7.9	DB	-34.1	31.2	2.4	1.7	-35.8	32.5	-37.2	33.5	-38.6	34.4	-40.4	35.7	(4)
(5)			WB	-34.2	21.7	2.4	1.2	-35.9	22.5	-37.3	23.2	-38.7	23.9	-40.4	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-10.6	-10.8	-4.8	3.6	10.6	14.6	16.9	15.5	10.9	3.9	-4.9	-11.0	(6)	
	DBStd	11.95	8.49	7.97	7.12	5.22	4.40	2.99	2.81	3.21	4.50	4.82	7.03	7.92	(7)	
	HDD10.0	3268	638	583	460	198	46	3	0	2	42	196	448	651	(8)	
	HDD18.3	5671	897	817	718	443	241	117	61	96	226	448	698	909	(9)	
	CDD10.0	674	0	0	0	6	65	141	215	174	68	6	0	0	(10)	
	CDD18.3	34	0	0	0	0	1	4	18	9	2	0	0	0	(11)	
	CDH23.3	629	0	0	0	1	49	81	247	169	82	1	0	0	(12)	
	CDH26.7	123	0	0	0	0	6	9	57	31	20	0	0	0	(13)	
(14) Wind	WSAvg	3.8	3.9	3.6	3.9	4.3	4.2	3.8	3.4	3.3	3.6	3.9	3.7	3.7	(14)	
(15) Precipitation	PrecAvg	413	19	11	17	30	44	70	81	54	33	19	19	16	(15)	
	PrecMax	574	69	23	37	65	118	128	139	103	82	44	77	39	(16)	
	PrecMin	221	3	1	6	2	7	21	30	11	4	1	1	3	(17)	
	PrecStd	73	15	6	8	16	22	28	27	19	21	11	16	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.9	11.5	22.0	27.7	28.0	31.0	30.0	29.6	21.8	12.6	6.8	(19)	
		MCWB	2.7	2.8	5.8	10.3	13.8	16.6	19.1	17.4	15.7	11.6	6.8	3.0	(20)	
	2%	DB	4.5	4.2	7.7	18.3	24.6	25.7	28.2	27.0	25.7	16.7	8.2	3.9	(21)	
		MCWB	1.5	1.1	3.6	8.7	12.4	15.3	18.8	17.2	14.4	9.3	4.0	1.0	(22)	
	5%	DB	2.9	2.2	5.6	15.1	22.0	23.7	26.0	25.2	22.5	13.7	5.5	1.9	(23)	
		MCWB	0.6	0.0	2.3	7.0	11.4	14.8	17.8	16.8	13.4	7.3	2.3	-0.3	(24)	
	10%	DB	1.0	0.2	3.7	12.3	19.6	21.7	24.1	23.2	19.1	11.3	3.4	-0.2	(25)	
		MCWB	-0.8	-1.5	1.0	5.5	10.2	13.7	17.0	16.0	11.9	6.3	1.1	-1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.1	6.4	10.8	14.6	18.6	21.6	20.2	16.6	12.1	7.0	3.2	(27)	
		MCDB	5.7	6.3	10.5	21.1	24.9	25.2	27.5	25.6	25.5	19.8	11.7	6.2	(28)	
	2%	WB	1.8	1.4	4.0	9.0	13.2	16.8	19.8	18.7	15.1	9.8	4.5	1.3	(29)	
		MCDB	4.1	3.8	7.2	17.7	22.1	22.6	25.8	24.6	24.2	15.3	7.8	3.7	(30)	
	5%	WB	0.7	0.0	2.4	7.3	12.1	15.7	18.6	17.5	13.8	8.2	2.7	-0.2	(31)	
		MCDB	2.8	2.0	5.2	14.0	20.3	21.6	24.8	23.3	21.3	12.7	5.1	1.8	(32)	
	10%	WB	-0.9	-1.6	1.2	5.9	11.0	14.7	17.5	16.4	12.5	6.8	1.2	-1.9	(33)	
		MCDB	1.0	0.3	3.4	11.3	18.2	20.1	23.1	22.1	18.2	10.8	3.1	-0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	9.2	10.8	12.6	11.5	11.9	12.4	12.3	10.5	8.4	8.4	(35)	
		MCDBR	9.4	11.6	10.9	16.5	16.2	14.8	15.3	16.1	18.2	14.9	10.7	10.6	(36)	
	5% WB	MCWBR	7.3	9.0	7.5	8.4	7.1	6.6	7.5	8.0	8.3	8.5	7.2	8.4	(37)	
		MCWBR	9.4	10.8	10.3	15.3	14.9	12.5	14.0	14.4	16.4	14.1	10.2	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.259	0.306	0.330	0.345	0.360	0.358	0.369	0.308	0.276	0.251	0.241	(40)	
		taud	2.303	2.350	2.316	2.364	2.377	2.358	2.394	2.333	2.488	2.549	2.443	2.280	(41)	
	Ebn at Noon		737	863	880	902	903	888	884	850	875	840	752	675	(42)	
		Edh at Noon	57	75	98	107	112	116	110	110	83	62	49	47	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.97	2.02	3.56	4.68	5.75	6.01	6.16	5.06	3.51	2.02	0.97	0.72	(44)	
	RadStd		0.11	0.15	0.28	0.44	0.42	0.40	0.31	0.35	0.37	0.25	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page